was reused. The route every one of these bodies of data takes, from the run that produced it to the tensors a model is fitted on, is shown in Figure 4.1, and the sections below follow it in order.

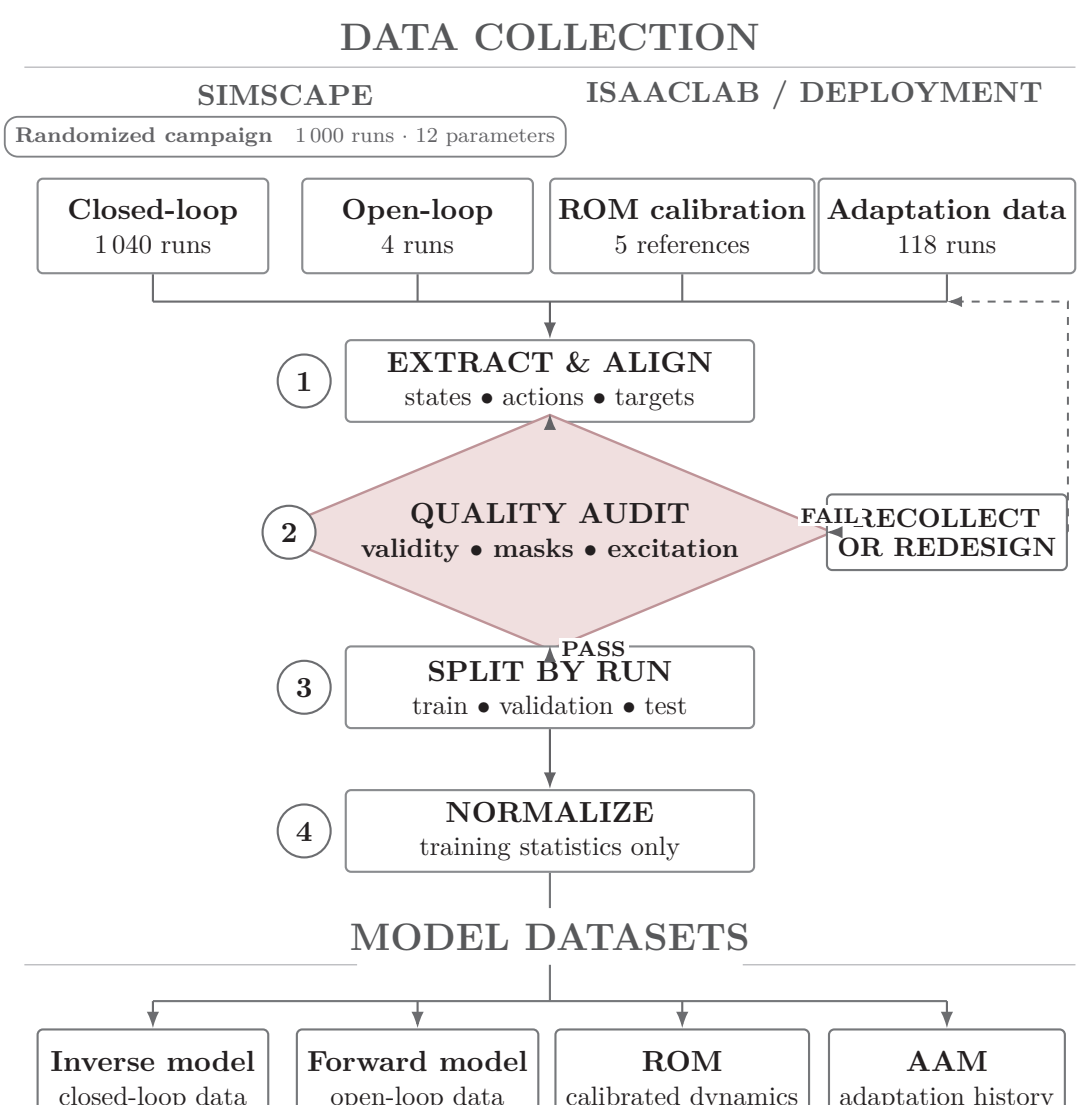



**Figure 4.1:** Data-preparation workflow. Two Simscape collections and two IsaacLab collections pass through common extraction, audit, run-level splitting and training-statistics normalization stages. The randomized Simscape campaign is provenance metadata rather than a pipeline stage. A failed audit returns the experiment to collection; accepted material forms the inverse, forward, ROM and AAM datasets.

## 4.1 Dataset Objectives and Design Requirements

Before any run was collected I fixed four requirements that the data had to satisfy jointly. They are stated explicitly, because the rest of the chapter is essentially the record of discovering that no single collection strategy meets all four.

**R1: realistic operating distribution.** The learned models are deployed inside a tracking controller, so they must be accurate in the states that controller visits: coordinated